

Homework 3

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Question 1:

1. $\Rightarrow f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex function

$$\Rightarrow A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$$

$$\Rightarrow h(x) = f(Ax + b)$$

$$\Rightarrow \text{let } x_1, x_2 \in \mathbb{R}^n \text{ and } \theta \in [0, 1]$$

$$\Rightarrow h(\theta x_1 + (1-\theta)x_2) =$$

$$= f(A(\theta x_1 + (1-\theta)x_2) + b)$$

$$= f(\theta(Ax_1 + b) + (1-\theta)(Ax_2 + b))$$

$$\leq \theta f(Ax_1 + b) + (1-\theta)f(Ax_2 + b)$$

$$[\text{By the property of convexity of } f, \forall u, v \in \mathbb{R}^n \\ f(\theta u + (1-\theta)v) \leq \theta f(u) + (1-\theta)f(v)]$$

$$\Rightarrow h(\theta x_1 + (1-\theta)x_2) \leq \theta h(x_1) + (1-\theta)h(x_2)$$

$\Rightarrow$ We have used the property of affine nature of mapping $x \rightarrow Ax + b$ which preserves convex combinations

$$\Rightarrow \boxed{h(x) \text{ is convex function}}$$

2. $\Rightarrow f: \mathbb{R}^n \rightarrow \mathbb{R}, g: \mathbb{R}^n \rightarrow \mathbb{R}$ are convex functions

$$\Rightarrow A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$$

$$\Rightarrow k(x) = \max \{f(x), g(x)\}$$

$$\Rightarrow x_1, x_2 \in \mathbb{R}^n, \theta \in [0, 1]$$

$$\Rightarrow k(\theta x_1 + (1-\theta)x_2)$$

$$= \max \{f(\theta x_1 + (1-\theta)x_2), g(\theta x_1 + (1-\theta)x_2)\}$$

$$\Rightarrow f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2)$$

($\because f$ is convex)

$$\Rightarrow f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2)$$

$$\leq \theta \max\{f(x_1), g(x_1)\}$$

$$\Rightarrow g(\theta x_1 + (1-\theta)x_2) \leq \theta g(x_1) + (1-\theta)g(x_2) \quad \text{--- ①}$$

$$\leq \theta \max\{f(x_1), g(x_1)\} + (1-\theta) \max\{f(x_2), g(x_2)\}$$

($\because g$ is convex)

--- ②

$$\Rightarrow k(\theta x_1 + (1-\theta)x_2)$$

$$= \max\{f(\theta x_1 + (1-\theta)x_2), g(\theta x_1 + (1-\theta)x_2)\}$$

$\Rightarrow$ By ① and ②,

$$k(\theta x_1 + (1-\theta)x_2) \leq \theta k(x_1) + (1-\theta)k(x_2)$$

$k(x)$ is convex

$$\Rightarrow m(x) = \min\{f(x), g(x)\}$$

$\Rightarrow$ Take $f(x) = x$, $g(x) = -x$ on $\mathbb{R}$.

Both $f(x)$ and $g(x)$ are convex functions.

$$m(x) = \min\{x, -x\} = -|x|$$

For $m(x)$ to be convex, $x_1, x_2 \in \mathbb{R}$ and $\theta \in [0, 1]$

$$m(\theta x_1 + (1-\theta)x_2) \leq \theta m(x_1) + (1-\theta)m(x_2)$$

Take counterexample construction of $x_1 = -1, x_2 = 1$,

$$\theta = \frac{1}{2}$$

$$m(\theta x_1 + (1-\theta)x_2) = m\left(\frac{x_1 + x_2}{2}\right) = m(0) = 0$$

$$\theta m(x_1) + (1-\theta)m(x_2) = \left(\frac{1}{2}\right)m(-1) + \frac{1}{2}m(1) = \frac{-1}{2} + \frac{-1}{2} = -1$$

$$m(\theta x_1 + (1-\theta)x_2) > \theta m(x_1) + (1-\theta)m(x_2)$$

$m(x)$ is not necessarily convex

Question 2:

$\Rightarrow S^n$ denotes space of real symmetric $n \times n$ matrices

$\Rightarrow S_+^n = \{X \in S^n \mid X \succeq 0\}$ is the cone of PSD matrices

$\Rightarrow S_+^n$ is a proper cone in S^n (fact to be used)

$\Rightarrow$ Define generalized inequality (non-strict and strict) induced by this cone.

$\Rightarrow$ Cone K is called proper cone if it is:

① Convex: $\forall x, y \in K, \theta \in [0, 1]; \theta x + (1-\theta)y \in K$.

② Closed: has all limit points

③ Solid: $\text{int } K \neq \emptyset$

④ Pointed: $\text{if } x \in K \text{ and } -x \in K \text{ then } x = 0$

$\Rightarrow X \preceq_K Y \iff Y - X \in K$

$X \prec_K Y \iff Y - X \in \text{int } K$

$\Rightarrow$ Now, here $K = S_+^n$

Non-strict inequality:

$\Rightarrow M \in S_+^n \iff \forall v \in \mathbb{R}^n, v^T M v \geq 0$

$M \in S_+^n \iff M \text{ is PSD}$

$\Rightarrow X \preceq_K Y \iff Y - X \in K \text{ i.e. } Y - X \in S_+^n$

$\iff v^T (Y - X) v \geq 0 \forall v \in \mathbb{R}^n$

$\iff \boxed{Y - X \text{ is PSD matrix}} \iff Y - X \in S_+^n$

Strict inequality:

$\Rightarrow X \prec_{S_+^n} Y \text{ means } Y - X \in \text{int}(S_+^n)$

$\Rightarrow$ We need to show that $\text{int}(S_+^n) = S_{++}^n$

$\Rightarrow$ Then $X \prec_{S_+^n} Y \iff Y - X \in \text{int}(S_+^n)$

$\iff Y - X \in S_{++}^n \iff Y - X \succ 0$
($Y - X$ is PD)

$$\iff v^T(Y-X)v > 0 \quad \forall v \in \mathbb{R}^n, v \neq 0, (Y-X \text{ is positive definite})$$

$\Rightarrow$ Now, problem reduces to proving $\text{int}(S_+^n) = S_{++}^n$

$\Rightarrow$ If M is PD, $\exists \epsilon > 0$, $\epsilon = \lambda_{\min}(M)$ such that

$\forall D \in S_+^n$ with $\|D\|_2 < \epsilon$ (spectral norm),

$M+D$ is PD. (Weyl's inequality)

$\Rightarrow B_\epsilon(M)$ is contained in $S_+^n \iff M \in \mathcal{O}$

$\Rightarrow$ If M is PD, $M \in \text{int}(S_+^n)$

$\Rightarrow$ If M is on boundary (singular), then $\forall \epsilon > 0$, we can make some small change to push it out of cone

$\Rightarrow \text{int}(S_+^n) = S_{++}^n$

$\Rightarrow M \in S_+^n$ is on boundary iff $|M| = 0$ i.e. $\exists v \neq 0$ such that $v^T M v = 0$. M is interior point iff it is invertible and PD.

$\Rightarrow$ Role of cone's interior in defining strict inequality:

$\Rightarrow$ Interior of cone is important for strict ordering.

$\Rightarrow$ If $X \leq_K Y$, $Y-X \in \text{bd } S_+^n$ is also possible. Boundary matrix is singular. Atleast 1 eigen value is 0.

$$v^T(Y-X)v = 0 \rightarrow \exists v \text{ that satisfies this}$$

$\Rightarrow$ If $X \prec_K Y$, then $Y-X \in \text{int}(S_+^n)$. For an interior point, it can be surrounded by small open ball for any interior point. ~~Ball~~ Ball still there in cone for any interior point.

$\Rightarrow$ If $Y-X \in \text{int}(S_+^n)$, even if $Y-X$ is perturbed by very small symmetric matrix, it will be PSD.

$\Rightarrow \lambda_{\min}(Y-X) > 0$ for interior point. So, $v^T(Y-X)v > 0 \quad \forall v \in \mathbb{R}^n \setminus 0$. $Y-X$ is Positive Definite

Question 3:

1. $\Rightarrow K \subset \mathbb{R}^n$. $K = \{c \in \mathbb{R}^n \mid c_1 + c_2 t + \dots + c_n t^{n-1} \geq 0 \forall t \in [0, 1]\}$

$\Rightarrow$ Prove that K is a proper cone.

$\Rightarrow$ Proper cone is convex cone, closed and pointed, and ~~has~~ solid (non empty interior).

~~$\Rightarrow$~~ K is cone:

$\Rightarrow$ Let $c \in K$ and $\theta \geq 0$. For every $t \in [0, 1]$:

$$c_1 + c_2 t + \dots + c_n t^{n-1} \geq 0.$$

$$\Rightarrow \theta c = \theta c_1 + \theta c_2 t + \dots + \theta c_n t^{n-1}$$

$$\theta c \geq 0 \quad \forall t \in [0, 1]. \quad \theta c \in K$$

K is cone

K is convex:

$\Rightarrow$ Let $c, d \in K$ and $\theta \in [0, 1]$. For every $t \in [0, 1]$

$$\Rightarrow \theta c_1 + (1-\theta)d_1 + (\theta c_2 + (1-\theta)d_2)t + \dots + (\theta c_n + (1-\theta)d_n)t^{n-1}$$

$$= \theta (c_1 + c_2 t + \dots + c_n t^{n-1}) + (1-\theta)(d_1 + d_2 t + \dots + d_n t^{n-1})$$

$$\geq 0$$

$$\Rightarrow \theta c + (1-\theta)d \in K.$$

K is convex

K is closed:

$\Rightarrow$ For each fixed $t \in [0, 1]$, $f_t: \mathbb{R}^n \rightarrow \mathbb{R}$

$$f_t(c) = c_1 + c_2 t + \dots + c_n t^{n-1}$$

$\hookrightarrow$ linear, continuous

$f_t^{-1}([0, \infty))$ is closed,

$$K = \bigcap_{t \in [0, 1]} f_t^{-1}([0, \infty))$$

Intersection of closed sets is closed. K is closed

Pointed: $\Rightarrow \nexists c \in K$ and $-c \in K$. For every $t \in [0, 1]$

$$\Rightarrow c_1 + c_2 t + \dots + c_n t^{n-1} \geq 0$$

$$-(c_1 + c_2 t + \dots + c_n t^{n-1}) \geq 0 \rightarrow \text{Finite degree of } n-1$$

$$\Rightarrow p(t) = c_1 + c_2 t + \dots + c_n t^{n-1}$$

$$\Rightarrow p(t) = 0 \quad \forall t \in [0, 1].$$

so at most $n-1$ roots but ∞ roots into $[0, 1]$.

~~$\Rightarrow$ Non-zero~~ $\Rightarrow K$ is pointed

Solid: $\Rightarrow$ Take $c = (1, 0, 0, \dots, 0)$

$$\Rightarrow p_c(t) = 1 > 0 \quad \forall t \in [0, 1].$$

$\Rightarrow [0, 1]$ is compact. $\exists$ constant M such that

for any d with $\|d - c\| < \delta$, $(t^{i-1} \leq 1 \text{ on } t \in [0, 1])$

$$|p_d(t) - 1| \leq \sum_{i=1}^n |d_i - c_i| t^{i-1}$$

$$\leq \|d - c\| \sum_{i=1}^n 1 \leq n \|d - c\|$$

$\delta < \frac{1}{n} \rightarrow p_c(t) > 0 \quad \forall t. K$ has non empty interior.

K is solid.

$\Rightarrow K$ is a proper cone

2. $\Rightarrow$ Generalized inequality $c \leq_K d$ induced by cone K

$$\Rightarrow K \subset \mathbb{R}^n; K = \{c \in \mathbb{R}^n \mid c_1 + c_2 t + \dots + c_n t^{n-1} \geq 0 \quad \forall t \in [0, 1]\}$$

$\Rightarrow \forall c, d \in \mathbb{R}^n, c \leq_K d$ means $d - c \in K$

$$(d_1 - c_1) + (d_2 - c_2)t + \dots + (d_n - c_n)t^{n-1} \geq 0 \quad \forall t \in [0, 1]$$

$$c_1 + c_2 t + \dots + c_n t^{n-1} \leq d_1 + d_2 t + \dots + d_n t^{n-1} \quad \forall t \in [0, 1]$$

$$\Rightarrow c \leq_K d \text{ iff } c_1 + c_2 t + \dots + c_n t^{n-1}$$

$$\leq d_1 + d_2 t + \dots + d_n t^{n-1} \quad \forall t \in [0, 1]$$

Question 4

⇒ Binary classification problem with linearly separable dataset $\{(x_i, y_i)_{i=1}^n\}$ where $x_i \in \mathbb{R}^n$, $y_i \in \{-1, 1\}$

1. ⇒ linear classifier $f(x) = w^T x + b$, $w \in \mathbb{R}^n$, $b \in \mathbb{R}$

⇒ Data point (x_i, y_i) , $y_i \in \{-1, 1\}$

⇒ With respect to this datapoint

⇒ Functional margin = $y_i (w^T x_i + b)$

This measures how correctly and confidently point is classified. Positive value indicates correct classification. Larger magnitude means more confidence. Negative value indicates incorrect classification

⇒ Geometric margin = $\frac{y_i (w^T x_i + b)}{\|w\|}$

⊗ This is signed Euclidean distance from point to classifier. It is invariant to scaling of (w, b) .

2. ⇒ Show that maximizing the minimum geometric margin over all training samples is equivalent to solving optimization problem

$\min_{w, b} \frac{1}{2} \|w\|_2^2$ subject to $y_i (w^T x_i + b) \geq 1 \quad \forall i$

⇒ $\max_{w, b} \min_i \frac{y_i (w^T x_i + b)}{\|w\|}$

⇒ let M be the minimum geometric margin

⇒ $\max_{w, b} M$ subject to $\frac{y_i (w^T x_i + b)}{\|w\|} \geq M \quad \forall i$

⇒ $\max_{w, b} M$ subject to $y_i (w^T x_i + b) \geq \|w\| M \quad \forall i$

⇒ Fix this $\|w\| M = 1$, (it is just scaling)
of (w, b)

$$\Rightarrow \text{let } K = \min_i y_i (w^T x_i + b), \quad K > 0$$

↳ linearly separable data

$$\Rightarrow M = \frac{K}{\|w\|}$$

$$\Rightarrow \text{let } w' = \frac{w}{K}, \quad b' = \frac{b}{K}.$$

$$\Rightarrow y_i (w'^T x_i + b') = \frac{y_i (w^T x_i + b)}{K} \geq 1$$

$$\Rightarrow \frac{y_i (w^T x_i + b)}{K} = 1 \text{ for minimum margin.}$$

$$\Rightarrow M = \frac{K}{\|w\|} = \frac{1}{\|w'\|}$$

$$\Rightarrow \text{maximize } M \longrightarrow \text{minimize } \|w'\|$$

$$\longrightarrow \text{minimize } \frac{1}{2} \|w'\|^2 \longrightarrow \text{minimize } \|w\|$$

$$\Rightarrow \min_{w, b} \frac{1}{2} \|w\|^2 \text{ subject to } y_i (w^T x_i + b) \geq 1 \quad \forall i$$

This optimization problem can be solved (hard-margin SVM primal problem)